



HISTORY

NETWORK PROJECT



김동진 이영훈 장규혁 정성현



TABLE OF CONTENTS

01 논리적 구성도

01-1 IP 구성도

01-2 IGP 구성도

02 물리적 구성도

02-1 물리적 구성도

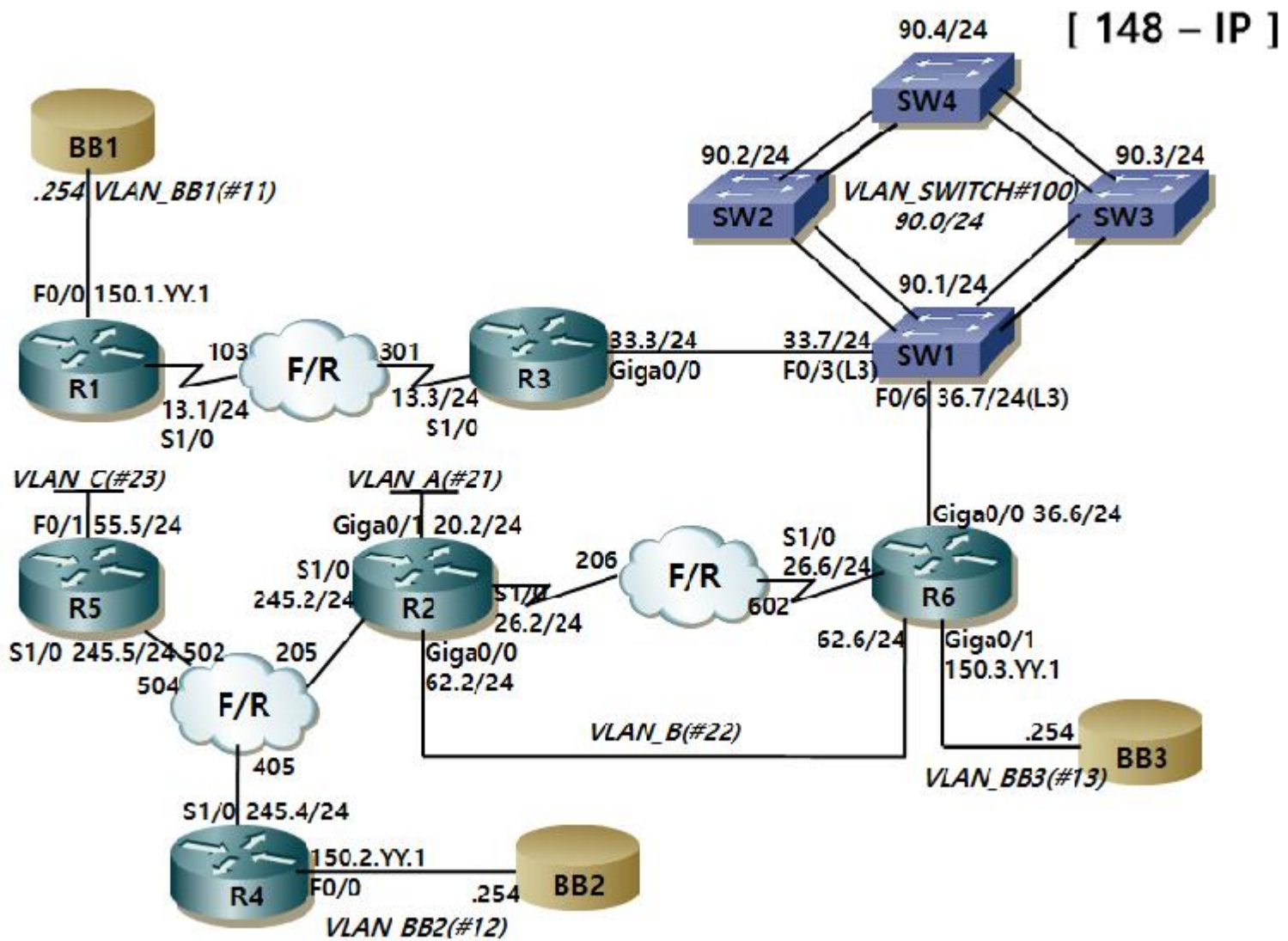
02-2 전제

03 설정

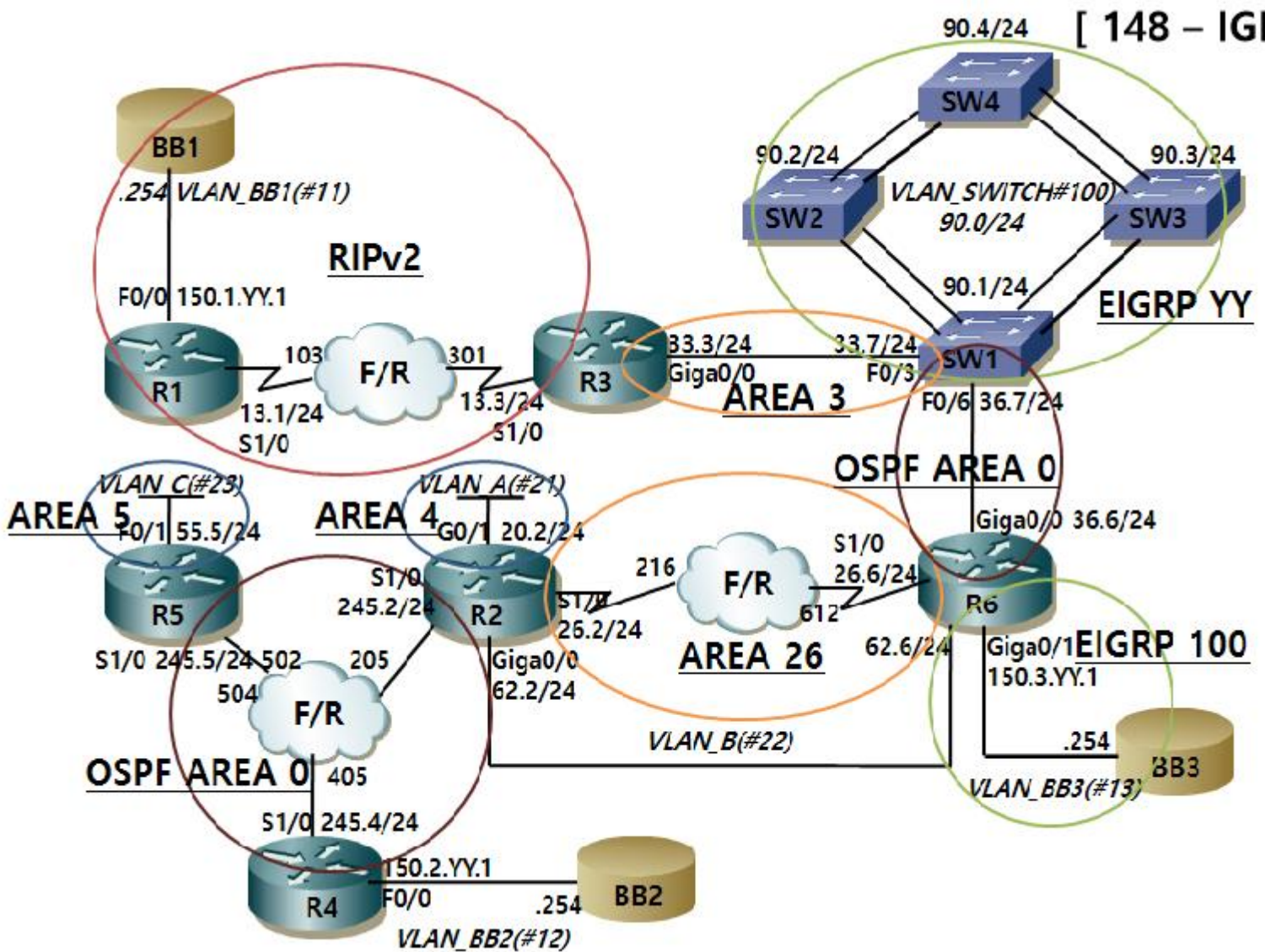
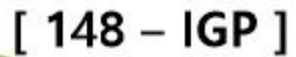
03-1 결과

논리적 구성도

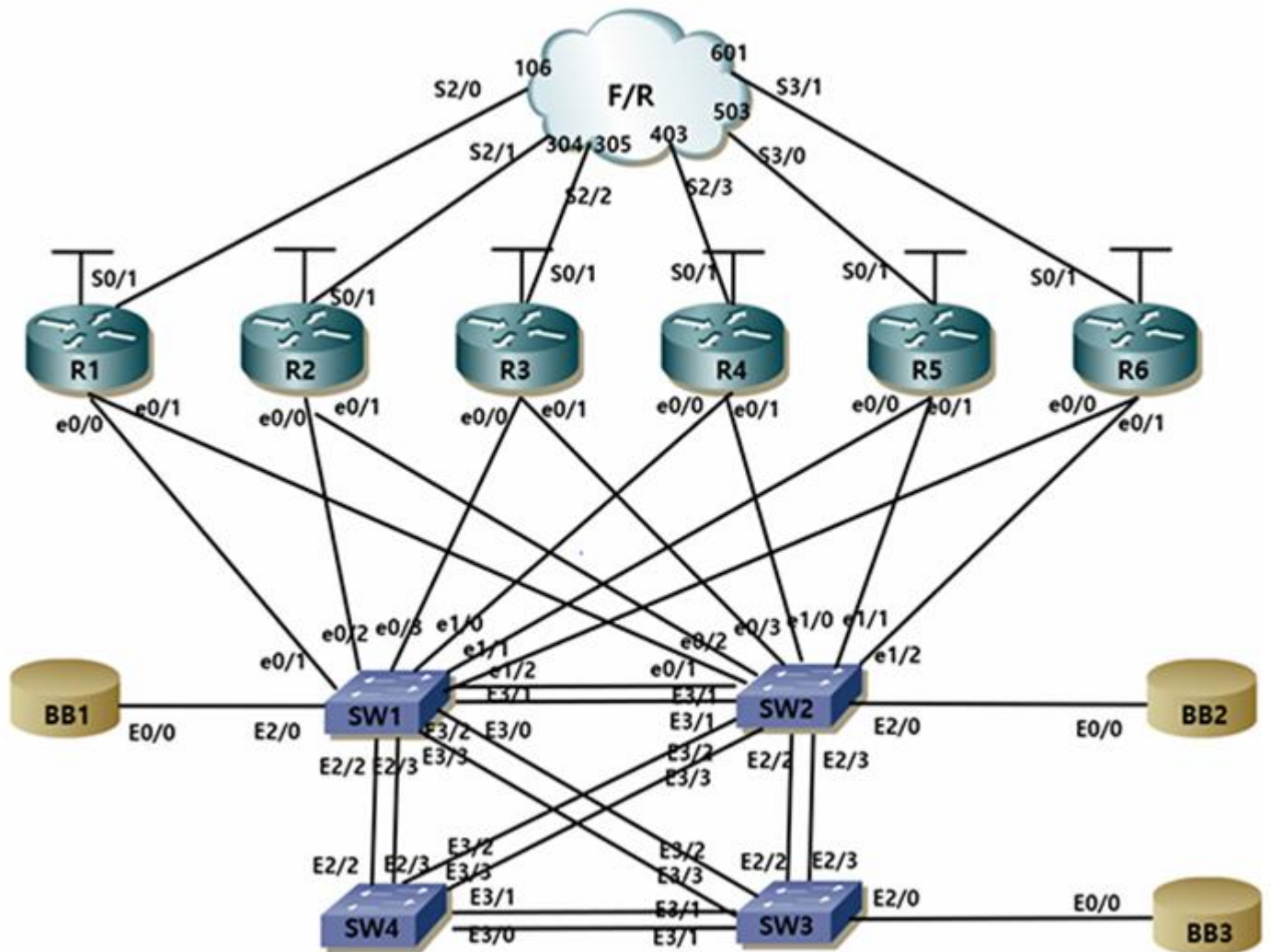
IP



IGP



물리적 구성도



Address Allocation

General Information

IOS 12.4

Doc CD : you have access to cisco.com/univercd

All configuration guides and master indexes are there

Tools : notepad and calculator are available

0. Address Allocation

use class B address range YY.YY.X.0/16. YY is your rack number

YY is your rack number, X is your router number

Ex (If your Rack number is 03, Lo0 is 3.3.3.3 and Rack number is 07, Lo0 is 7.7.7.7)

Note

- VLAN_A (#21) YY.20.0/24
 - VLAN_B (#22) YY.62.0/24
 - VLAN_C (#23) YY.55.0/24
 - VLAN_CUSTOMER1(CUSTOMER_A) (#50)
 - VLAN_CUSTOMER2(CUSTOMER_B) (#79)
 - VLAN_SWITCHES (#100) YY.90.0/24
 - VLAN_BB1 (#11)
 - VLAN_BB2 (#12)
 - VLAN_BB3 (#13)
 - FrameRelay : YY.YY.13.0/24 (R1-R3)
YY.YY.26.0/30(YY.62.0/24) (R2-R6)
YY.YY.245.0/24(YY.24.0/24) (R2-R4-R5)
 - BB1 is 150.1.YY.254/24
 - BB2 is 150.2.YY.254/24
 - BB3 is 150.3.YY.254/24
- <loopback interface0 /24> (YY.YY.X.X : X=router number)

T/S

Frame-relay 관련 설정 미리 되어 있으며, 별도의 그림이 제공된다.

SW의 VTP mode 는 Transparent mode 로 설정 되어 있다.

SW에 vlan 설정되어 있는 부분은 없다. 문제에서 vlan 설정을 요구함.

EIGRP 구간에 있는 VLAN의 SVI interface 와 Loopback address는 설정되어 있지 않음. 문제에서 요구

R3의 G0/1 에 사용하지 않는 IP address 가 설정되어 있음(삭제)

R2,R6 serial network 와 R2,R6의 Ethernet network 가 바뀔 수 있음

Bridging and Switching

1. Configure IP across your frame relay network

frame-relay 구성 후 자기 자신의 IP address 로 ping 이 되게 설정하시오.

R1

```
int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.103 point-to-point
ip address 14.14.13.1 255.255.255.0
frame-relay interface-dlci 103
```

R3

```
int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.301 point-to-point
ip address 14.14.13.3 255.255.255.0
frame-relay interface-dlci 301
```

R2

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.245 multipoint
ip address 14.14.245.2 255.255.255.0
frame-relay map ip 14.14.245.5 205 broadcast
frame-relay map ip 14.14.245.4 205 broadcast
frame-relay map ip 14.14.245.2 205
ip ospf network broadcast

interface s1/0.206 point-to-point
ip address 14.14.26.2 255.255.255.0
frame-relay interface-dlci 206
```

R4

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

int s1/0.245 multipoint
ip add 14.14.245.4 255.255.255.0
frame-relay map ip 14.14.245.5 405 broadcast
frame-relay map ip 14.14.245.2 405 broadcast
frame-relay map ip 14.14.245.4 405
ip ospf network broadcast
```

R6

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.602 point-to-point
ip address 14.14.26.6 255.255.255.0
frame-relay interface-dlci 602
```

R5

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

int s1/0.245 multipoint
ip add 14.14.245.5 255.255.255.0
frame-relay map ip 14.14.245.2 502 broadcast
frame-relay map ip 14.14.245.4 504 broadcast
frame-relay map ip 14.14.245.5 502
ip ospf network broadcast
```

Bridging and Switching

2.1 VTP Configuration

configure so that sw2, sw3 and sw4 will learn vlan information from sw1

SW1

vtpp mode server
vtpp domain history.com
vtpp password history
vtpp version 2

SW2

vtpp mode client
vtpp domain history.com
vtpp password history
vtpp version 2

SW3

vtpp mode client
vtpp domain history.com
vtpp password history
vtpp version 2

SW4

vtpp mode client
vtpp domain history.com
vtpp password history
vtpp version 2

SW1 결과

```
VTP Version capable      : 1 to 3
VTP version running      : 2
VTP Domain Name         : history.com
VTP Pruning Mode        : Disabled
VTP Traps Generation     : Disabled
Device ID                : aabb.cc00.0000
Configuration last modified by 0.0.0.0 at 2-26-26 06:29:57
Local updater ID is 14.14.99.1 on interface V1100 (lowest numbered VLAN interface found)

Feature VLAN:
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 14
Configuration Revision   : 31
MD5 digest               : 0x3D 0xAC 0x96 0x10 0x7F 0x0F 0x5C 0x4D
                        : 0xDC 0x8C 0x85 0x83 0x72 0x86 0xC6 0x5C
```

2.2 Trunk Port

R6-SW2 사이를 Trunk port 로 만들고, 필요한 VLAN 정보만 흘러 다니도록 설정하라

R6

interface e0/1
no shutdown

interface e0/1.13
encapsulation dot1q 13
ip address 150.3.14.1 255.255.255.0

interface e0/1.22
encapsulation dot1q 22
ip address 14.14.62.6 255.255.255.0

SW2

interface e1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 13,22

Bridging and Switching

2.3 SW1,SW2,SW3,SW4

SW1-SW4

interface range e2/2-3

shutdown

SW1

interface range e3/0-1

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 12 mode on

interface range e3/2-3

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 13 mode on

SW2

interface range e3/0-1

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 12 mode on

interface range e3/2-3

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 24 mode on

SW3

interface range e3/0-1

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 34 mode on

interface range e3/2-3

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 13 mode on

SW4

interface range e3/0-1

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 34 mode on

interface range e3/2-3

switchport trunk encapsulation dot1q

switchport mode trunk

channel-group 24 mode on

SW1 결과

Group	Port-channel	Protocol	Ports	
-----+-----+-----+-----				
12	Po12(SU)	-	Et3/0(P)	Et3/1(P)
13	Po13(SU)	-	Et3/2(P)	Et3/3(P)

2.4 Etherchannel Loadbalancing

SW1-SW2

port-channel load-balance src-dst-ip

Bridging and Switching

2.5 VLAN configuration

SW1

```
vlan 11
name BB1
vlan 12
name BB2
vlan 13
name BB3
vlan 21
name VLAN_A
vlan 22
name VLAN_B
vlan 23
name VLAN_C
vlan 50
name VLAN_CUSTOMER1
vlan 79
name VLAN_CUSTOMER2
vlan 100
name VLAN_SWITCHS

int e2/0
switchport mode access
switchport access vlan 11

int e0/0
switchport mode access
switchport access vlan 11

int e0/2
switch port mode access
switchport access vlan 22

interface e0/3
no switchport
ip address 14.14.33.7 255.255.255.0

interface e1/2
no switchport
ip address 14.14.36.7 255.255.255.0
```

SW2

```
interface e2/0
switchport mode access
switchport access vlan 12

interface e1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 13,22

interface e0/2
switchport mode access
switchport access vlan 21

int e1/1
switchport mode access
switchport access vlan 23
```

SW3

```
interface e2/0
switchport mode access
switchport access vlan 13
```

Bridging and Switching

SW1

```
SW1#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	Fi0/1, Fi1/0, Fi1/1, Fi1/3 Ei2/1
11	BB1	active	Ei0/0, Ei2/0
12	BB2	active	
13	BB3	active	
21	VLAN_A	active	
22	VLAN_B	active	Ei0/2
23	VLAN_C	active	
50	VLAN_CUSTOMER1	active	
79	VLAN_CUSTOMER2	active	
100	VLAN_SWITCH	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

SW2

VLAN	Name	Status	Ports
1	default	active	Fr0/0, Fr0/1, Fr0/3, Fr1/0 Ei1/3, Ei2/1
11	BB1	active	
12	BB2	active	Fr2/0
13	BB3	active	
21	VLAN_A	active	Li0/2
22	VLAN_B	active	
23	VLAN_C	active	Ei1/1
50	VLAN_CUSTOMER1	active	
79	VLAN_CUSTOMER2	active	
100	VLAN_SWITCH	active	
1002	fddi-default	act/unsup	
1003	trcrf default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

Bridging and Switching

SW3

VLAN	Name	Status	Ports
1	default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/0, Et1/1, Et1/2, Et1/3 Et2/1
11	BB1	active	
12	BB2	active	
13	BB3	active	Et2/0
21	VLAN_A	active	
22	VLAN_B	active	
23	VLAN_C	active	
50	VLAN_CUSTOMER1	active	
79	VLAN_CUSTOMER2	active	
100	VLAN_SWITCH	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

SW4

VLAN	Name	Status	Ports
1	default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/0, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1
11	BB1	active	
12	BB2	active	
13	BB3	active	
21	VLAN_A	active	
22	VLAN_B	active	
23	VLAN_C	active	
50	VLAN_CUSTOMER1	active	
79	VLAN_CUSTOMER2	active	
100	VLAN_SWITCH	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

Bridging and Switching

2.6 STP shared vlan

SW1

```
spanning-tree mode mst
spanning-tree mst configuration
name HISTORY
revision 1
instance 1 vlan 11,21
instance 2 vlan 100
spanning-tree mst 1 root primary
```

SW4

```
spanning-tree mode mst
spanning-tree mst configuration
name HISTORY
revision 1
instance 1 vlan 11,21
instance 2 vlan 100
spanning-tree mst 2 root primary
```

SW3

```
spanning-tree mode mst
spanning-tree mst configuration
name HISTORY
revision 1
instance 1 vlan 11,21
instance 2 vlan 100
```

SW2

```
spanning-tree mode mst
spanning-tree mst configuration
name HISTORY
revision 1
instance 1 vlan 11,21
instance 2 vlan 100
```

결과

SW1 - mst

```
SW1#show span mst conf
Name      [HISTORY]
Revision  1      Instances configured 3

Instance  Vlan mapped
-----
0         1-10,12-20,22-99,101-4094
1         11,21
2         100
-----
```

Bridging and Switching

SW1 - root

```
SW1#show span vlan 11

MST1
Spanning tree enabled protocol mstp
Root ID    Priority    24577
           Address    aabb.cc00.0700
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
           Address    aabb.cc00.0700
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Et0/0          Desg FWD 2000000   128.1   Shr
Et2/0          Desg FWD 2000000   128.65  Shr
Po13           Desg FWD 1000000   128.514 Shr
Po12           Desg FWD 1000000   128.515 Shr
```

SW4 - root

```
SW4#show span vlan 100

MST2
Spanning tree enabled protocol mstp
Root ID    Priority    24578
           Address    aabb.cc00.0a00
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    24578 (priority 24576 sys-id-ext 2)
           Address    aabb.cc00.0a00
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Po24           Desg BLK 1000000   128.514 Shr Dispute
Po34           Desg BLK 1000000   128.515 Shr Dispute
```


Bridging and Switching

2.8 UDLD

SW3-SW4

```
int ran e3/0-1
udld port aggressive
```

2.10 Mac Address Database

SW3

```
mac address-table aging-time 500 vlan 13
```

결과

```
Global Aging Time: 300
Vlan      Aging Time
-----
13        500
```

2.11 Root Switch

SW1-SW4

```
spanning-tree mst configuration
instance 3 vlan 12
```

SW2

```
spanning-tree mst 3 root primary
```

SW1- mst

```
SW1#show span mst conf
Name      [HISTORY]
Revision  1      Instances configured 4

Instance  Vlans mapped
-----
0         1-10,13-20,22-99,101-4094
1         11,21
2         100
3         12
```

Bridging and Switching

SW2 - root

```
SW2#show span vlan 12

MST3
Spanning tree enabled protocol mstp
Root ID    Priority    24579
           Address    aabb.cc00.0800
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    24579 (priority 24576 sys-id-ext 3)
           Address    aabb.cc00.0800
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface   Role Sts Cost      Prio.Nbr Type
-----
Et2/0       Desg FWD 2000000   128.65  Shr
Po24        Desg LRN 1000000   128.514 Shr
Po12        Desg FWD 1000000   128.515 Shr
```

2.13 Create the interfaces VLAN below VLAN 100 (VLAN_SWITCHES)

SW1

```
interface vlan 100
no shutdown
ip address 14.14.90.1 255.255.255.0
```

SW2

```
interface vlan 100
no shutdown
ip address 14.14.90.2 255.255.255.0
```

SW3

```
interface vlan 100
no shutdown
ip address 14.14.90.3 255.255.255.0
```

SW4

```
interface vlan 100
no shutdown
ip address 14.14.90.4 255.255.255.0
```

결과

SW1

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/3	14.14.33.7	YES	manual	up	up
Ethernet1/2	14.14.36.7	YES	manual	up	up
Loopback0	14.14.7.7	YES	manual	up	up
Vlan100	14.14.90.1	YES	manual	up	up

SW2

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback0	14.14.90.2	YES	manual	up	up

SW3

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback0	14.14.9.9	YES	manual	up	up
Vlan100	14.14.90.3	YES	manual	up	up

SW4

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback0	14.14.10.10	YES	manual	up	up
Vlan100	14.14.90.4	YES	manual	up	up

IP IGP Protocols

1.1 RIPv2 Configuration

R1

```
router rip
version 2
no auto-summary
network 150.1.0.0
network 14.0.0.0
passive-interface default
neighbor 150.1.14.254
neighbor 14.14.13.3
```

R3

```
router rip
version 2
no auto-summary
network 14.0.0.0
passive-interface default
neighbor 14.14.13.1
```

2. EIGRP

2.1 EIGRP YY Configuration

SW1

```
ip routing
router eigrp 14
no auto-summary
network 14.14.90.1 0.0.0.0
```

SW2

```
ip routing
router eigrp 14
no auto-summary
network 14.14.8.8 0.0.0.0
network 14.14.90.2 0.0.0.0
```

SW3

```
ip routing
router eigrp 14
no auto-summary
network 14.14.9.9 0.0.0.0
network 14.14.90.3 0.0.0.0
```

SW4

```
ip routing
router eigrp 14
no auto-summary
network 14.14.10.10 0.0.0.0
network 14.14.90.4 0.0.0.0
```

2.2 EIGRP 100 configuration

R6

```
router eigrp 100
no auto-summary
network 150.3.14.1 0.0.0.0
eigrp stub redistribute
```

IP IGP Protocols

3. OSPF

3.1 AREA 0

R2

```
router ospf 14
net 14.14.245.2 0.0.0.0 area 0
int s1/0.245
ip ospf priority 0
```

R4

```
router ospf 14
net 14.14.245.4 0.0.0.0 area 0
net 14.14.4.4 0.0.0.0 area 0
int s1/0.245
ip ospf priority 0
```

R5

```
router ospf 14
net 14.14.245.5 0.0.0.0 area 0
neighbor 14.14.245.2
neighobr 14.14.245.4
```

SW1

```
router ospf 14
network 14.14.36.7 0.0.0.0 area 0
```

R6

```
router ospf 14
network 14.14.36.6 0.0.0.0 area 0
```

3.2 AREA 26 (virtual-link)

R2

```
router ospf 14
net 14.14.62.2 0.0.0.0 area 26
net 14.14.26.2 0.0.0.0 area 26
area 26 virtual-link 14.14.6.6

int s1/0.206
ip ospf network point-to-point
int e0/0
ip ospf network point-to-point
```

R6

```
router ospf 14
network 14.14.62.6 0.0.0.0 area 26
network 14.14.26.6 0.0.0.0 area 26
network 14.14.6.6 0.0.0.0 area 26
area 26 virtual-link 14.14.2.2

int s1/0.602
ip ospf network point-to-point
int e0/1.22
ip ospf network point-to-point
```

IP IGP Protocols

II. IP IGP Protocols

3.3 AREA 3

R3

```
router ospf 14
network 14.14.3.3 0.0.0.0 area 3
network 14.14.33.3 0.0.0.0 area 3
```

SW1

```
router ospf 14
network 14.14.33.7 0.0.0.0 area 3
network 14.14.7.7 0.0.0.0 area 3
```

3.4 AREA 4

R2

```
router ospf 14
net 14.14.2.2 0.0.0.0 area 4
net 14.14.20.2 0.0.0.0 area 4
area 4 stub no-summary
```

3.5 AREA 5

R5

```
router ospf 14
net 14.14.55.5 0.0.0.0 area 5
net 14.14.5.5 0.0.0.0 area 5
area 5 stub
```

4. Redistribution

4.1 RIP – OSPF

R3

```
router rip
redistribute ospf 14 metric 3

router ospf 14
redistribute rip subnets
```

4.2 OSPF – EIGRP YY

SW1

```
router eigrp 14
redistribute ospf 14 metric 1 1 1 1 1

router ospf 14
redistribute eigrp 14 subnets
```

IP IGP Protocols

4.3 OSPF – EIGRP 100

```

R6

router ospf 14
redistribute eigrp 100 subnets

router eigrp 100
redistribute ospf 14 metric 1 1 1 1 1
eigrp stub redistribute

```

5. Loopback 어드레스는 라우팅 테이블에 /32로 보여지면 안된다.(ip ospf network point-to-point)

```

R1-R6

interface lo0
ip ospf network point-to-point

```

```

SW1-SW4

interface lo0
ip ospf network point-to-point

```

결과

Routing / Redistribute

SW1 Routing

```

14.0.0.0/8 is variably subnetted, 18 subnets, 2 masks
O IA 14.14.2.2/32 [110/21] via 14.14.36.6, 00:02:52, Ethernet1/2
O    14.14.3.0/24 [110/11] via 14.14.33.3, 00:02:17, Ethernet0/3
O IA 14.14.6.0/24 [110/11] via 14.14.36.6, 00:02:52, Ethernet1/2
C    14.14.7.0/24 is directly connected, Loopback0
L    14.14.7.7/32 is directly connected, Loopback0
D    14.14.8.0/24 [90/130816] via 14.14.90.2, 00:02:55, Vlan100
D    14.14.9.0/24 [90/130816] via 14.14.90.3, 00:02:50, Vlan100
D    14.14.10.0/24 [90/130816] via 14.14.90.4, 00:02:37, Vlan100
O IA 14.14.20.0/24 [110/30] via 14.14.36.6, 00:02:52, Ethernet1/2
O IA 14.14.26.0/24 [110/74] via 14.14.36.6, 00:02:52, Ethernet1/2
C    14.14.33.0/24 is directly connected, Ethernet0/3
L    14.14.33.7/32 is directly connected, Ethernet0/3
C    14.14.36.0/24 is directly connected, Ethernet1/2
L    14.14.36.7/32 is directly connected, Ethernet1/2
O IA 14.14.62.0/24 [110/20] via 14.14.36.6, 00:02:52, Ethernet1/2
C    14.14.90.0/24 is directly connected, Vlan100
L    14.14.90.1/32 is directly connected, Vlan100
O    14.14.245.0/24 [110/84] via 14.14.36.6, 00:02:52, Ethernet1/2

```


IP IGP Protocols

R3 Routing

```

14.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
R      14.14.1.0/24 [120/1] via 14.14.13.1, 00:00:14, Serial1/0.301
O IA   14.14.2.2/32 [110/31] via 14.14.33.7, 00:00:24, Ethernet0/0
C      14.14.3.0/24 is directly connected, Loopback0
L      14.14.3.3/32 is directly connected, Loopback0
O IA   14.14.6.0/24 [110/21] via 14.14.33.7, 00:00:24, Ethernet0/0
O      14.14.7.0/24 [110/11] via 14.14.33.7, 00:00:24, Ethernet0/0
C      14.14.13.0/24 is directly connected, Serial1/0.301
L      14.14.13.3/32 is directly connected, Serial1/0.301
O IA   14.14.20.0/24 [110/40] via 14.14.33.7, 00:00:24, Ethernet0/0
O IA   14.14.26.0/24 [110/84] via 14.14.33.7, 00:00:24, Ethernet0/0
C      14.14.33.0/24 is directly connected, Ethernet0/0
L      14.14.33.3/32 is directly connected, Ethernet0/0
O IA   14.14.36.0/24 [110/20] via 14.14.33.7, 00:00:24, Ethernet0/0
O IA   14.14.62.0/24 [110/30] via 14.14.33.7, 00:00:24, Ethernet0/0
O IA   14.14.245.0/24 [110/94] via 14.14.33.7, 00:00:24, Ethernet0/0
150.1.0.0/24 is subnetted, 1 subnets
R      150.1.14.0 [120/1] via 14.14.13.1, 00:00:14, Serial1/0.301

```

R6 Routing

```

14.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
O IA   14.14.2.2/32 [110/11] via 14.14.62.2, 00:03:13, Ethernet0/1.22
O IA   14.14.3.0/24 [110/21] via 14.14.36.7, 00:01:35, Ethernet0/0
C      14.14.6.0/24 is directly connected, Loopback0
L      14.14.6.6/32 is directly connected, Loopback0
O IA   14.14.7.0/24 [110/11] via 14.14.36.7, 00:02:10, Ethernet0/0
O IA   14.14.20.0/24 [110/20] via 14.14.62.2, 00:03:13, Ethernet0/1.22
C      14.14.26.0/24 is directly connected, Serial1/0.602
L      14.14.26.6/32 is directly connected, Serial1/0.602
O IA   14.14.33.0/24 [110/20] via 14.14.36.7, 00:02:10, Ethernet0/0
C      14.14.36.0/24 is directly connected, Ethernet0/0
L      14.14.36.6/32 is directly connected, Ethernet0/0
C      14.14.62.0/24 is directly connected, Ethernet0/1.22
L      14.14.62.6/32 is directly connected, Ethernet0/1.22
O      14.14.245.0/24 [110/74] via 14.14.62.2, 00:03:14, Ethernet0/1.22
150.3.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      150.3.14.0/24 is directly connected, Ethernet0/1.13
L      150.3.14.1/32 is directly connected, Ethernet0/1.13

```


IP IGP Protocols

R1 Redistribute

```

14.0.0.0/8 is variably subnetted, 21 subnets, 2 masks
C    14.14.1.0/24 is directly connected, Loopback0
L    14.14.1.1/32 is directly connected, Loopback0
R    14.14.2.2/32 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.3.0/24 [120/1] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.4.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.5.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.6.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.7.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.8.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.9.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R    14.14.10.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
C    14.14.13.0/24 is directly connected, Serial1/0.103
L    14.14.13.1/32 is directly connected, Serial1/0.103
R    14.14.20.0/24 [120/1] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.26.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.33.0/24 [120/1] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.36.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.55.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.62.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.90.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R    14.14.245.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
150.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    150.1.14.0/24 is directly connected, Ethernet0/0
L    150.1.14.1/32 is directly connected, Ethernet0/0
150.3.0.0/24 is subnetted, 1 subnets
R    150.3.14.0 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103

```

R5 Redistribute

```

14.0.0.0/8 is variably subnetted, 22 subnets, 2 masks
O E2    14.14.1.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA    14.14.2.2/32 [110/65] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA    14.14.3.0/24 [110/95] via 14.14.245.2, 00:01:44, Serial1/0.245
O       14.14.4.0/24 [110/65] via 14.14.245.4, 00:01:54, Serial1/0.245
C       14.14.5.0/24 is directly connected, Loopback0
L       14.14.5.5/32 is directly connected, Loopback0
O IA    14.14.6.0/24 [110/75] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA    14.14.7.0/24 [110/85] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2    14.14.8.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2    14.14.9.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2    14.14.10.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2    14.14.13.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA    14.14.20.0/24 [110/74] via 14.14.245.2, 00:01:45, Serial1/0.245
O IA    14.14.26.0/24 [110/128] via 14.14.245.2, 00:01:45, Serial1/0.245
O IA    14.14.33.0/24 [110/94] via 14.14.245.2, 00:01:45, Serial1/0.245
O       14.14.36.0/24 [110/84] via 14.14.245.2, 00:01:45, Serial1/0.245
C       14.14.55.0/24 is directly connected, Ethernet0/1
L       14.14.55.5/32 is directly connected, Ethernet0/1
O IA    14.14.62.0/24 [110/74] via 14.14.245.2, 00:01:45, Serial1/0.245
O E2    14.14.90.0/24 [110/20] via 14.14.245.2, 00:01:45, Serial1/0.245
C       14.14.245.0/24 is directly connected, Serial1/0.245
L       14.14.245.5/32 is directly connected, Serial1/0.245
150.1.0.0/24 is subnetted, 1 subnets
O E2    150.1.14.0 [110/20] via 14.14.245.2, 00:01:45, Serial1/0.245
150.3.0.0/24 is subnetted, 1 subnets
O E2    150.3.14.0 [110/20] via 14.14.245.2, 00:01:45, Serial1/0.245

```

IOS Feature / Security

2. DHCP

R6

```
ip dhcp excluded-address 14.14.62.6
ip dhcp excluded-address 14.14.62.2
ip dhcp pool DHCP
network 14.14.62.0 255.255.255.0
dns-server 150.100.1.50 150.100.1.51
domain-name history.com
lease infinite
default-router 14.14.62.6 14.14.62.2
```

3. UDP Broadcast Management

R6

```
ip forward-protocol udp bootpc
int e0/1.13
ip helper-address 150.2.14.244
```

1. OSPF Authentication

R2

```
router ospf 14
area 0 authentication message-digest
area 26 virtual-link 14.14.6.6 message-digest-key 1
md5 history
int s1/0
ip ospf message-digest-key 1 md5 history
```

R6

```
router ospf 14
area 0 authentication message-digest
area 26 virtual-link 14.14.2.2 message-digest-key 1
md5 history
interface e0/0
ip ospf message-digest-key 1 md5 history
```

R4

```
router ospf 14
area 0 authentication message-digest
int s1/0
ip ospf message-digest-key 1 md5 history
int lo0
ip ospf message-digest-key 1 md5 history
```

R5

```
router ospf 14
area 0 authentication message-digest
int s1/0
ip ospf message-digest-key 1 md5 history
```

SW1

```
router ospf 14
area 0 authentication message-digest
int e1/2
ip ospf message-digest-key 1 md5 history
```



HISTORY
